

1. Given a matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, recall that one can calculate its eigenvalues by solving the characteristic polynomial $\lambda^2 - (a + d)\lambda + (ad - bc) = 0$. In this quiz, you will practice calculating and solving the characteristic polynomial to find the eigenvalues of simple matrices.

1 point

For the matrix $A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$, what is the characteristic polynomial, and the solutions to the characteristic polynomial?

- ☐ $\lambda^2 - 3\lambda - 2 = 0$
 $\lambda_1 = 1, \lambda_2 = -2$
- ☒ $\lambda^2 - 3\lambda + 2 = 0$
 $\lambda_1 = 1, \lambda_2 = 2$
- ☐ $\lambda^2 + 3\lambda + 2 = 0$
 $\lambda_1 = -1, \lambda_2 = -2$
- ☐ $\lambda^2 + 3\lambda - 2 = 0$
 $\lambda_1 = -1, \lambda_2 = 2$

2. Recall that for a matrix A , the eigenvectors of the matrix are vectors for which applying the matrix transformation is the same as scaling by some constant.

1 point

For $A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$ as immediately above, select all eigenvectors of this matrix.

- ☒ $\begin{bmatrix} 0 \\ 3 \end{bmatrix}$
- ☒ $\begin{bmatrix} 0 \\ 2 \end{bmatrix}$
- ☐ $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$
- ☒ $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$

3. For the matrix $A = \begin{bmatrix} 3 & 4 \\ 0 & 5 \end{bmatrix}$, what is the characteristic polynomial, and the solutions to the characteristic polynomial?

1 point

- ☒ $\lambda^2 - 8\lambda + 15 = 0$
 $\lambda_1 = 3, \lambda_2 = 5$
- ☐ $\lambda^2 + 8\lambda - 15 = 0$
 $\lambda_1 = 3, \lambda_2 = -5$
- ☐ $\lambda^2 + 8\lambda + 15 = 0$
 $\lambda_1 = -3, \lambda_2 = -5$
- ☐ $\lambda^2 - 8\lambda - 15 = 0$
 $\lambda_1 = -3, \lambda_2 = 5$

4. For the matrix $A = \begin{bmatrix} 3 & 4 \\ 0 & 5 \end{bmatrix}$ as immediately above, select all eigenvectors of this matrix.

1 point

- ☐ $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$
- ☒ $\begin{bmatrix} 3 \\ 0 \end{bmatrix}$
- ☒ $\begin{bmatrix} -1 \\ -1/2 \end{bmatrix}$
- ☒ $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$

5. For the matrix $A = \begin{bmatrix} 1 & 0 \\ -1 & 4 \end{bmatrix}$, what is the characteristic polynomial, and the solutions to the characteristic polynomial?

1 point

- ☒ $\lambda^2 - 5\lambda + 4 = 0$
 $\lambda_1 = 1, \lambda_2 = 4$

$$\lambda_1 = 1, \lambda_2 = -4$$

☐ $\lambda^2 + 5\lambda - 4 = 0$

$$\lambda_1 = 1, \lambda_2 = -4$$

☐ $\lambda^2 - 5\lambda - 4 = 0$

$$\lambda_1 = -1, \lambda_2 = 4$$

☐ $\lambda^2 + 5\lambda + 4 = 0$

$$\lambda_1 = -1, \lambda_2 = -4$$

6. For the matrix $A = \begin{bmatrix} 1 & 0 \\ -1 & 4 \end{bmatrix}$ as immediately above, select all eigenvectors of this matrix.

1 point

☐ $\begin{bmatrix} 3 \\ -1 \end{bmatrix}$

☒ $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$

☒ $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$

☐ $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$

7. For the matrix $A = \begin{bmatrix} -3 & 8 \\ 2 & 3 \end{bmatrix}$, what is the characteristic polynomial, and the solutions to the characteristic polynomial?

1 point

☐ $\lambda^2 + 25 = 0$

$$\lambda_1 = \lambda_2 = -5$$

☐ $\lambda^2 - 25 = 0$

$$\lambda_1 = \lambda_2 = 5$$

☐ $\lambda^2 + 25 = 0$

$$\lambda_1 = -5, \lambda_2 = 5$$

☒ $\lambda^2 - 25 = 0$

$$\lambda_1 = -5, \lambda_2 = 5$$

8. For the matrix $A = \begin{bmatrix} -3 & 8 \\ 2 & 3 \end{bmatrix}$ as immediately above, select all eigenvectors of this matrix.

1 point

☒ $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

☒ $\begin{bmatrix} -1 \\ -1 \end{bmatrix}$

☐ $\begin{bmatrix} 0 \\ 2 \end{bmatrix}$

☒ $\begin{bmatrix} 4 \\ -1 \end{bmatrix}$

9. For the matrix $A = \begin{bmatrix} 5 & 4 \\ -4 & -3 \end{bmatrix}$, what is the characteristic polynomial, and the solutions to the characteristic polynomial?

1 point

☒ $\lambda^2 - 2\lambda + 1 = 0$

$$\lambda_1 = \lambda_2 = 1$$

☐ $\lambda^2 - 2\lambda + 1 = 0$

$$\lambda_1 = -1, \lambda_2 = 1$$

☐ $\lambda^2 - 2\lambda + 1 = 0$

$$\lambda_1 = \lambda_2 = -1$$

☐ $\lambda^2 - 2\lambda + 1 = 0$

No real solutions.

10. For the matrix $A = \begin{bmatrix} -2 & -3 \\ 1 & 1 \end{bmatrix}$, what is the characteristic polynomial, and the solutions to the characteristic polynomial?

1 point

☐ $\lambda^2 - \lambda + 1 = 0$

No real solutions.

☐ $\lambda^2 + \lambda - 1 = 0$

$$\lambda_1 = \frac{-\sqrt{5}-1}{2}, \lambda_2 = \frac{\sqrt{5}-1}{2}$$

☐ $\lambda^2 - \lambda - 1 = 0$

$$\lambda_1 = \frac{1-\sqrt{5}}{2}, \lambda_2 = \frac{1+\sqrt{5}}{2}$$

☒ $\lambda^2 + \lambda + 1 = 0$

No real solutions.